IT4010 Research Project – 2026 Jan

Topic Assessment Form

V2.1

Project ID: **R26-IT-135**

1. Topic (12 words max)

Offline-First Communication Platform for Outdoor and Event Management

2. Research group the project belongs to

SST - Software Systems & Technologies

3. Specialization of the project belongs to

Information Technology (IT)

4. If a continuation of a previous project:

Project ID	-
Year	-

5. Brief description of the research problem including references (300 words max) – references not included in word count.

Reliable communication systems in outdoor and remote areas have always been a challenge for hikers, event organizers, and even emergency services. In numerous instances, such as hiking trails, outdoor events, or disaster-struck zones with little to no cellular connectivity, the availability and reliability of communication systems have been an issue. The intended users for the proposed system include hikers, event organizers with a small number of event attendees, and even the respective authorities in the event location who may be in need of offline communication services for connectivity in remote locations.

The consequences of such communication deficiencies are diverse. Socially, the lack of real-time communication may pose a threat to personal security and group synchronization. Technologically, the lack of messaging platforms when communication goes offline forces a reliance on cellular networks and the internet. Economically, poor communication, either delayed or missed, may cause inefficient resource allocation and/or higher operational expenses during events/emergencies. Ethical/LEGAL considerations include the paramount need to guarantee secure communication for protecting user privacy. Environmentally, the heavier use of traditional infrastructure equates to more power consumption, as compared to the power-consuming, loaded-lite mesh communication supporting local events.

This research falls within the Communication and Networking cluster, with potential vertical industries such as mobile communication, disaster management, or outdoor recreation services. The proposed platform will help maintain sustainable and resilient communication networks, thus helping the United Nations attain its sustainable development goals through enabling safe, reliable, and low-cost communication without dependence on conventional infrastructure: SDG 9 (Industry, Innovation, and Infrastructure), SDG 11 (Sustainable Cities and Communities), and SDG 3 (Good Health and Well-being).

The system can further be improved by deploying transmission repeater devices in strategic locations along trails or event areas to extend message reach and reliability even further. These repeaters will work with the mobile application to keep offline multi-hop routing with increased overall network robustness.

References

- Bluemergency: Mediating Post-disaster Communication Systems using the Internet of Things and Bluetooth Mesh (2019)
- arXiv: https://arxiv.org/abs/1909.08094
- IEEE Xplore (conference version): https://ieeexplore.ieee.org/document/9033063
- Survey on Bluetooth Low Energy Mesh Networks (including emergency applications like HIRO-NET for disaster search-and-rescue): https://pmc.ncbi.nlm.nih.gov/articles/PMC7349184/
- Overview of Bluetooth Low Energy Mesh applications, with explicit mention of suitability for emergency communication ("Bluemergency" system): https://pmc.ncbi.nlm.nih.gov/articles/PMC9965677/

6. Brief description of Existing Research and Systems (300 words max)

Current communication systems, like those in Whatsapp, Telegram, and Signal, support end-to-end, real-time messaging delivery and acknowledgement, but these systems are highly dependent on internet or cellular networks and, thus, are inefficient in areas without infrastructure or during disasters. Non-internet, offline messaging applications, GoTenna and Bridgefy, have used Bluetooth or mesh technology for internet-independent communication; however, these applications have faced issues related to range, scalabilit, and appropriate routing for long-distance message delivery. Research papers on Delay Tolerant Networks (DTNs) and Mobile Ad-hoc Networks (MANETs) have concentrated on opportunistic and SCF routing for improving connectivity in disadvantaged areas for message delivery. However, most present systems and concepts have several limitations, namely short-range communication without a means for long-distance delivery, no delivery confirmations similar to Whatsapp, no prioritization for emergency messages, and little attention paid for usability on mobile devices.

Moreover, there are no such networks which use optional transmission repeater devices for optimal coverage in outdoor or event networks. These examples reveal an essential need for an effective offline first communication solution which can facilitate short range direct messaging, decentralized multi hop routing, tolerance for delays in transport, in addition to context-aware management.

7. Brief description of the solution's nature, including a conceptual diagram (maximum 500 words).

The solution entitled “Offline-First Communication Platform for Outdoor and Event Management” offers a solution to the identified shortcomings of the current systems by means of an effective, secure, and accessible offline communication solution. Unlike most online communication systems like messengers that rely on the use of cellular or internet connectivity, this system will provide communication services in areas where the infrastructure is lacking or unavailable by means of hikers, event organizers, or during emergencies. The solution will utilize direct communication, routing through the software, delay tolerance for messages, as well as software-controlled communication innovation to ensure the messages are delivered to the target audience efficiently, securely, and safely. Additional transmission repeater kits may also be utilized to bridge the existing solution gaps within sparsely populated zones.

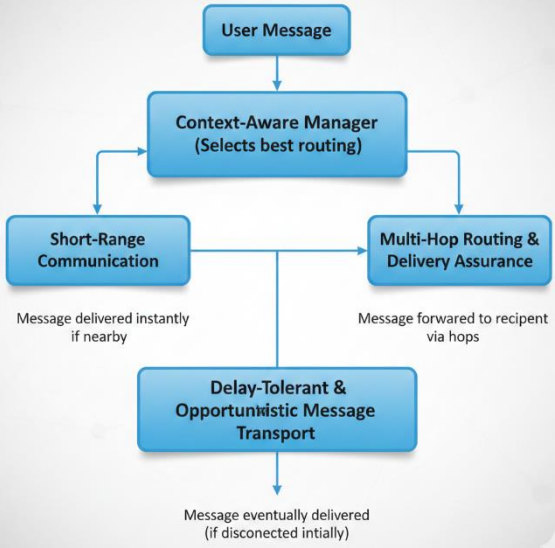
System perspective: “Input, Process, and Output” can be considered a solution to the problem at a system perspective in the following way:

Input: The message is composed by users using a mobile application. Every message contains message content, target users, priority level, normal or emergency, and timestamp. Additionally, information on neighboring devices, connectivity, as well as battery conditions of devices is constantly acquired to facilitate routing.

Process: For the optimal transmission method, the Context-Aware Communication Manager considers distance, connectivity, and the priority of the message. When the target individual is in close proximity, the Short-Range Direct Communication Layer uses Bluetooth BLE to send the message. However, when the target individual is outside the transmission distance, the Decentralized Multi-Hop Routing service is used to deliver the message indirectly via intermediate devices. Yet, when the target individual is outside the delivery distance of intermediate devices, the Delay-Tolerant and Opportunistic Message Transport service is used to send the message. All the while, message privacy is facilitated by message encryption.

Output: The system guarantees that messages reach their intended recipients either instantly (short-range) or eventually (long-distance or sparse network), with delivery status indicators provided to users. Group messages, emergency alerts, and event updates are handled seamlessly, while optional repeater devices enhance overall network coverage.

This design ensures **offline-first, scalable, secure, and user-friendly communication**, directly addressing the shortcomings of existing systems while providing practical utility for outdoor activities, event management, and emergency scenarios.



8. Brief overview of the availability of necessary specialized domain expertise, knowledge, and data. (500 words max)

The skill requirement for the proposed project, 'Offline-First Communication Platform for Outdoor and Event Management', includes knowledge of mobile application development, Bluetooth mesh communication, decentralized routing, delay tolerant networks, and simple data security methods. The team members of the project, being undergraduate students, have been studying for only the 4th year, and their required skill set could be developed through a blend of education and self-learning, along with the help of their project supervisor. The basic principles of networking, distributed systems, and mobile app development have been taught to the students as a part of their undergraduate studies.

For acquiring skills in the domain, one can resort to reading about the concepts of Bluetooth Low Energy mesh networks, opportunistic routing, and store-carry-forward delay tolerant networks. Open-source implementations of the concepts help in gaining better understanding of mesh networking, multi-hop routing, and context-aware communication management. Members can make the most of learning aids available in the form of e-learning portals, workshops, and publications in acquiring better concepts of lightweight encryption techniques and mobile application architecture pertaining to offline communication.

The skills related to the technological area that can be gained by working on this project include Android application development or cross-platform mobile application development skills, local database management skills for storing messages, as well as incorporation skills related to routing algorithms. Team members can also create simulations related to mesh networking by using a couple of devices.

The types of data necessary for the project include mainly message traffic for testing purposes, as well as data from the devices utilized by the users (IDs, proximity), as well as simulated outdoor conditions. The data may be obtained by conducting experiments using a variety of smartphones in indoor and outdoor settings, such as hiking trails, event locations, and emergencies. Test scenarios can be developed for a range of numbers and speeds for assessing the performance of message delivery. Other data necessary for the project will be generated dynamically.

Given that the system will collect only device IDs and message metadata for routing purposes, and the content of all messages is encrypted, ethical risks remain very limited. Ethical considerations, however, on user privacy and data security will be followed during the entire development and test process. If the trials require volunteers, informed consent will be sought, and identifiable information anonymized. The project does not involve sensitive personal data beyond encrypted messages; thus, formal ethical clearance may not be strictly required, though standard institutional guidelines will be followed.

This means that all the domain expertise, technical knowledge, and data that would be required are readily available to this team through a combination of coursework, online resources, experimentation, and supervised guidance in such a way as to make this project feasible, practical, and ethically responsible to implement by undergraduates.

9. Objectives and Novelty

Main Objective

To develop an offline communication system with the capability of providing trustworthy, secure, and friendly communication while engaging in outdoor activities, small events, or during emergency situations through the use of the Bluetooth mesh network, multi-hop routing, or delay-tolerant message transfer.

Member Name with Registration No	Sub Objective	Tasks	Novelty
IT22148704	Short-Range Direct Communication Layer	-Implement BLE-based direct messaging. -Enable one-to-one and group chat. -Implement message delivery status (✓ sent). -Integrate lightweight encryption for message privacy.	Offers offline messaging capabilities with delivery confirmations simply through BLE, which is suitable for outdoor/event usage. Delivery confirmations for offline mesh networks in real-time are rarely found in current implementations.
IT22168290	Decentralized Multi-Hop Routing & Delivery Assurance	- Implement multi-hop message forwarding logic. - Ensure messages reach intended recipient through intermediate nodes. - Include end-to-end delivery acknowledgments. - Test routing for small and medium-sized networks.	Adds WhatsApp-like delivery acknowledgement support for a decentralized offline network. Offers secure multi-hop forwarding support, which is less supported by existing solutions for an offline mesh network.

IT22544490	Context-Aware Communication Manager	- Develop decision-making logic to select best routing method. - Monitor device proximity, battery, and connectivity. - Prioritize emergency messages - Coordinate with all other components for smooth operation.	It applies lightweight, rule-based intelligence to perform offline routing and message prioritization, making efficient use of energy without relying on heavy AI.
IT22894038	Delay-Tolerant & Opportunistic Message Transport	- Implement local message storage for disconnected scenarios. - Forward stored messages opportunistically when devices come in range. - Integrate TTL (Time-To-Live) mechanism to prevent flooding. - Log delivery performance.	Coordinates delay tolerant networking technology, allowing data to be delivered over long distances even in offline mode, thereby enhancing the reliability of the existing offline apps.

10. This part is to be filled by the Supervisor and the Co-supervisor of the Project.

a) This research topic possesses a comprehensive scope suitable for a final-year project.

Yes	✓	No	

b) The proposed topic exhibits novelty.

Yes	✓	No	

c) The student group can successfully execute the proposed project.

Yes	✓	No	

d) The proposed sub-objectives reflect the students' areas of specialization.

Yes	✓	No	

e) Any other comments:

The scope of the project should be improved as advised. Accepting provided that the comments are addressed.

	Title	First Name	Last Name	Signature
Supervisor	Mr.	Vishan	Jayasinghearachchi	13/01/2026
Co-Supervisor				
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				